PHY202 Practice Exam 5 on chapter(s): 23, 24

1. $L = 4.39 \times 10^{-5}\,\mathrm{H} \because L = N\dfrac{\Delta\Phi}{\Delta t} = \dfrac{\mu_0 N^2 A}{\ell}$

2. $E_{ind} = 0.713\,\mathrm{J} \because E_{ind} = \dfrac{1}{2} L I^2$

3. $I = 4.46\,\mathrm{A} \because I(t) = I_0 e^{-t/\tau}, \tau = \dfrac{L}{R}$

4. $B_{max} = 6.00 \times 10^{-6}\,\mathrm{T} \because c = \dfrac{1}{\sqrt{\mu_0 \varepsilon_0}} = \dfrac{E}{B} = f\lambda$

5. $\mathrm{emf} = 3.07\,\mathrm{V} \because \mathrm{emf} = -N\dfrac{\Delta\Phi}{\Delta t}$

6. $\varepsilon = 106.\,\mathrm{V} \because \varepsilon = -N\dfrac{\Delta\Phi}{\Delta t}$

7. $\tau = 5.00 \times 10^{-3}\,\mathrm{s} \because \tau = \dfrac{L}{R}$

8. $I_p = 2.70\,\mathrm{A} \because \dfrac{V_s}{V_p} = \dfrac{N_s}{N_p} = \dfrac{I_p}{I_s}$

9. $\varepsilon = 2.92 \times 10^3\,\mathrm{V} \because \varepsilon = B\ell v$

10. $I_s = 2.55 \times 10^{-3}\,\mathrm{A} \because \dfrac{V_s}{V_p} = \dfrac{N_s}{N_p} = \dfrac{I_p}{I_s}$

11. $\varepsilon = 1.42 \times 10^3\,\mathrm{V} \because \varepsilon = NAB\omega \sin(\omega t), \sin(\omega t) = 1$

12. $E_0 = 1.97 \times 10^3\,\dfrac{\mathrm{V}}{\mathrm{m}} \because I_{ave} = \dfrac{c\,\varepsilon_0 E_0^2}{2} = \dfrac{c B_0^2}{2\mu_0} = \dfrac{E_0 B_0}{2\mu_0}$